

# CHRONIX

Evidence-Driven Incident Intelligence

## INCIDENT INVESTIGATION REPORT

### Payment API Incident

Incident ID: 5ead7581-eed3-466b-8b26-258bb9723c2e

Status: Active

| ROOT CAUSE | STATUS               |
|------------|----------------------|
| Assessment | <b>NOT CONFIRMED</b> |

Available evidence is insufficient or contradictory to establish a causal chain.

### Evidence to Reconstructed Events

| RAW EVIDENCE SIGNALS | > | RECONSTRUCTED TIMELINE EVENTS | > | TRACEABLE INCIDENT STORY |
|----------------------|---|-------------------------------|---|--------------------------|
| <b>7</b>             |   | <b>7</b>                      |   | Source-linked            |

Multiple evidence signals may support the same reconstructed timeline event.

### Executive Summary

Evidence reconstructed 7 event(s), beginning at 10:02. The timeline shows observable system changes and operational responses, but the available sources do not prove a single root cause.

The recorded sequence includes Payment API error rate increased (10:02); Database failure suspected (10:08); Payment service restarted (10:10); Database metrics returned to normal (10:12); Payment API error rate decreased (10:15).

Suspected explanations remain inferences; event order alone does not establish causation.

The available evidence establishes the recorded sequence, but is insufficient or contradictory to confirm a single root cause.

### Incident Timeline

| Time  | Event                               | Classification | Supporting evidence |
|-------|-------------------------------------|----------------|---------------------|
| 10:02 | Payment API error rate increased    | FACT           | demo-payment-001    |
| 10:03 | Database connection timeout         | FACT           | demo-payment-002    |
| 10:05 | Payment service deployment started  | FACT           | demo-payment-003    |
| 10:08 | Database failure suspected          | INFERENCE      | demo-payment-004    |
| 10:10 | Payment service restarted           | FACT           | demo-payment-005    |
| 10:12 | Database metrics returned to normal | FACT           | demo-payment-006    |

| Time  | Event                            | Classification | Supporting evidence |
|-------|----------------------------------|----------------|---------------------|
| 10:15 | Payment API error rate decreased | FACT           | demo-payment-007    |

## Classification Summary

| Timeline event classification | FACT | INFERENCE | CONFLICT | UNKNOWN |
|-------------------------------|------|-----------|----------|---------|
| Events                        | 6    | 1         | 0        | 0       |

The timeline has 7 events; its event classifications total 7. Facts and inferences are per-event classes. Conflicts and unknowns below are separate cross-event findings in the deterministic model.

| Cross-event analysis findings | Count |
|-------------------------------|-------|
| Conflicts                     | 1     |
| Unknowns                      | 1     |

## Response and Recovery

The timeline records Payment service restarted at 10:10. Recovery signals were observed afterward: Database metrics returned to normal (10:12); Payment API error rate decreased (10:15). The available evidence does not establish that the action directly caused recovery.

## Root Cause Assessment

| STATUS     | NOT CONFIRMED                                                                    |
|------------|----------------------------------------------------------------------------------|
| ASSESSMENT | Available evidence is insufficient or contradictory to establish a causal chain. |

## Established by Evidence

- 10:02 - Payment API error rate increased
- 10:03 - Database connection timeout
- 10:05 - Payment service deployment started
- 10:10 - Payment service restarted
- 10:12 - Database metrics returned to normal
- 10:15 - Payment API error rate decreased

## Not Established

- Inference only: Database failure suspected
- Causality cannot be established from the available evidence.

## Evidence Summary

Evidence received: **7 raw signals**. Reconstructed timeline: **7 events**.

Raw evidence can contain multiple signals that support one normalized timeline event. Event classifications and cross-event findings are reported separately above.

## Conflicts and Uncertainty

### ***Conflict findings***

- Evidence A (ChroniX Demo Payment Service): Database failure suspected. Evidence B (ChroniX Demo Payment Service): Database metrics returned to normal. Status: UNRESOLVED.

### ***Open unknowns***

- Causality cannot be established from the available evidence.

## Missing Evidence

The deterministic analysis records the items below as missing. Item-specific rationales are not stored; each gap relates to the open assessment: Causality cannot be established from the available evidence.

| Evidence needed            | Why it matters                                               |
|----------------------------|--------------------------------------------------------------|
| Database query/error logs  | Causality cannot be established from the available evidence. |
| Connection pool metrics    | Causality cannot be established from the available evidence. |
| Deployment impact analysis | Causality cannot be established from the available evidence. |

## Next Investigation Steps

No deterministic recommendations are recorded. Local-model suggestions are identified separately below as AI-generated guidance.

## Local AI Intelligence

Powered by local qwen3:4b via Ollama

AI-generated enrichment is hypothesis-only. Deterministic analysis remains authoritative.

**AI-generated interpretation:** Payment API error rate increased at 10:02, followed by database connection timeout at 10:03, payment service deployment started at 10:05, database failure suspected at 10:08, payment service restarted at 10:10, database metrics returned to normal at 10:12, and payment API error rate decreased at 10:15.

### ***Probable causes - AI hypothesis***

- Possible explanation suggested by AI (unconfirmed): Deployment of payment service 8f21 caused database connection issues. Direct causal evidence is not established. Evidence IDs: demo-payment-003.

### ***Contributing factors - AI interpretation***

- Potential contributing factor (not confirmed): Database connection timeout during payment processing Evidence IDs: demo-payment-002.

### ***Causal relationships - AI hypotheses***

- Unconfirmed AI causal hypothesis (correlation is not proof of causation): Deployment 8f21 may have introduced database connection issues. Deterministic root cause remains NOT CONFIRMED. Evidence IDs: demo-payment-003.

### ***Evidence interpretations - AI-generated***

- AI interpretation: Database failure suspected at 10:08 is consistent with database connection timeout at 10:03 Evidence IDs: demo-payment-002, demo-payment-004.

### ***Uncertainty - AI-generated***

- AI-identified uncertainty: Deployment impact analysis missing Evidence IDs: demo-payment-003.

### ***AI-generated recommendations***

- AI suggestion (investigator review required): Analyze deployment 8f21 logs for database connection issues

- AI suggestion (investigator review required): Review database query logs for error patterns

- AI suggestion (investigator review required): Check connection pool metrics during deployment

## **Evidence Traceability**

| Event                                       | Class     | Evidence IDs     | Source names                 |
|---------------------------------------------|-----------|------------------|------------------------------|
| 10:02 - Payment API error rate increased    | FACT      | demo-payment-001 | ChroniX Demo Payment Service |
| 10:03 - Database connection timeout         | FACT      | demo-payment-002 | ChroniX Demo Payment Service |
| 10:05 - Payment service deployment started  | FACT      | demo-payment-003 | ChroniX Demo Payment Service |
| 10:08 - Database failure suspected          | INFERENCE | demo-payment-004 | ChroniX Demo Payment Service |
| 10:10 - Payment service restarted           | FACT      | demo-payment-005 | ChroniX Demo Payment Service |
| 10:12 - Database metrics returned to normal | FACT      | demo-payment-006 | ChroniX Demo Payment Service |
| 10:15 - Payment API error rate decreased    | FACT      | demo-payment-007 | ChroniX Demo Payment Service |

## Appendix A - Raw Evidence

Raw source records are retained here for verification; the main report focuses on the reconstructed incident story.

| Evidence ID      | Source / type                              | Time / class                    | Raw evidence / metadata                                                                                            |
|------------------|--------------------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------|
| demo-payment-001 | ChroniX Demo Payment Service / application | 2026-09-25T10:02:00 / FACT      | HTTP 500 error rate exceeded threshold<br><b>Metadata:</b> severity: critical; service: payment-api                |
| demo-payment-002 | ChroniX Demo Payment Service / application | 2026-09-25T10:03:00 / FACT      | Payment database connection timeout observed<br><b>Metadata:</b> severity: critical; service: payment-api          |
| demo-payment-003 | ChroniX Demo Payment Service / application | 2026-09-25T10:05:00 / FACT      | Deployment rollout 8f21 started<br><b>Metadata:</b> severity: warning; service: payment-api                        |
| demo-payment-004 | ChroniX Demo Payment Service / application | 2026-09-25T10:08:00 / INFERENCE | Database failure may be contributing to payment errors<br><b>Metadata:</b> severity: warning; service: payment-api |
| demo-payment-005 | ChroniX Demo Payment Service / application | 2026-09-25T10:10:00 / FACT      | Payment service restarted during incident response<br><b>Metadata:</b> severity: critical; service: payment-api    |
| demo-payment-006 | ChroniX Demo Payment Service / application | 2026-09-25T10:12:00 / FACT      | Database CPU and connection metrics returned to normal<br><b>Metadata:</b> severity: info; service: payment-api    |
| demo-payment-007 | ChroniX Demo Payment Service / application | 2026-09-25T10:15:00 / FACT      | HTTP 500 error rate returned toward baseline<br><b>Metadata:</b> severity: info; service: payment-api              |

## Appendix B - Evidence Relationships

Identical persisted relationship rows are shown once. Correlation and inferred relationships are not confirmation of causality.

| Source ID        | Relationship    | Target ID        | Status and basis                                                                                                                                 |
|------------------|-----------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| demo-payment-001 | PRECEDED        | demo-payment-002 | deterministic: Chronological sequence: 'Payment API error rate increased' at 10:02:00 preceded 'Database connection timeout' at 10:03:00         |
| demo-payment-002 | PRECEDED        | demo-payment-003 | deterministic: Chronological sequence: 'Database connection timeout' at 10:03:00 preceded 'Payment service deployment started' at 10:05:00       |
| demo-payment-001 | CORRELATED_WITH | demo-payment-003 | deterministic: Correlated operational signals: payment                                                                                           |
| demo-payment-003 | PRECEDED        | demo-payment-004 | deterministic: Chronological sequence: 'Payment service deployment started' at 10:05:00 preceded 'Database failure suspected' at 10:08:00        |
| demo-payment-002 | CORRELATED_WITH | demo-payment-004 | deterministic: Correlated operational signals: database                                                                                          |
| demo-payment-004 | PRECEDED        | demo-payment-005 | deterministic: Chronological sequence: 'Database failure suspected' at 10:08:00 preceded 'Payment service restarted' at 10:10:00                 |
| demo-payment-001 | CORRELATED_WITH | demo-payment-005 | deterministic: Correlated operational signals: payment                                                                                           |
| demo-payment-003 | CORRELATED_WITH | demo-payment-005 | deterministic: Correlated operational signals: payment, service                                                                                  |
| demo-payment-005 | CONTRIBUTED_TO  | demo-payment-001 | deterministic: Remediation action: 'Payment service restarted' addressed prior degradation 'Payment API error rate increased'                    |
| demo-payment-005 | CONTRIBUTED_TO  | demo-payment-002 | deterministic: Remediation action: 'Payment service restarted' addressed prior degradation 'Database connection timeout'                         |
| demo-payment-005 | PRECEDED        | demo-payment-006 | deterministic: Chronological sequence: 'Payment service restarted' at 10:10:00 preceded 'Database metrics returned to normal' at 10:12:00        |
| demo-payment-002 | CORRELATED_WITH | demo-payment-006 | deterministic: Correlated operational signals: database                                                                                          |
| demo-payment-004 | CORRELATED_WITH | demo-payment-006 | deterministic: Correlated operational signals: database                                                                                          |
| demo-payment-004 | CONFLICTS_WITH  | demo-payment-006 | deterministic: Conflicting status between 'Database failure suspected' and 'Database metrics returned to normal'                                 |
| demo-payment-006 | PRECEDED        | demo-payment-007 | deterministic: Chronological sequence: 'Database metrics returned to normal' at 10:12:00 preceded 'Payment API error rate decreased' at 10:15:00 |
| demo-payment-001 | CORRELATED_WITH | demo-payment-007 | deterministic: Correlated operational signals: api, error, payment                                                                               |
| demo-payment-003 | CORRELATED_WITH | demo-payment-007 | deterministic: Correlated operational signals: payment                                                                                           |
| demo-payment-005 | CORRELATED_WITH | demo-payment-007 | deterministic: Correlated operational signals: payment                                                                                           |